

Note on the preservation of separability under species-resolved truncated SVD of the local PAW auxiliary

Internal note¹

¹Companion to paw_isdf_thc_prb.tex, Sec. V.A–V.D.
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This note clarifies a point raised about the derivation in Sec. V of the PAW-ISDF-THC manuscript: how the separability of the local auxiliary functions $\eta_{a\lambda}^{\mathbf{q}}(\mathbf{G}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \eta_{s(a)}(\lambda; \mathbf{q} + \mathbf{G})$ is preserved when the species-resolved factor is rotated to a truncated SVD basis. The conclusion is that separability is preserved by construction, because the rotation acts on an index that is absent from the atom phase, and the only approximation introduced by truncation is a rank reduction in the channel index. We also distinguish this exact algebraic statement from the empirical claim that the $\mathbf{q} = 0$ basis also covers η_s at $\mathbf{q} \neq 0$, which is a separate, evidence-based assertion.

I. WHAT “SEPARABILITY” MEANS IN THE ORIGINAL DEFINITION

The starting point is the species-resolved factorization of the local auxiliary, Eq. (eta-species-factor) of the manuscript,

$$\eta_{a\lambda}^{\mathbf{q}}(\mathbf{G}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \eta_{s(a)}(\lambda; \mathbf{q} + \mathbf{G}). \quad (1)$$

This is *not* a tensor-product decomposition over the indices $(\lambda, \mathbf{G})$: both factors carry a $\mathbf{G}$ dependence. The “separability” that the manuscript claims is structural and concerns the role of the atom index a :

- the atom dependence is carried entirely by the unitary phase $e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a}$, which has *no* channel index λ ;
- the channel/species dependence is carried entirely by $\eta_s(\lambda; \mathbf{q} + \mathbf{G})$, which depends on the atom only through $a \mapsto s(a)$.

This particular separation – atom phase factored out from a species function – is what enables a per-species (rather than per-atom) low-rank compression in the first place: a basis built once for species s applies, modulo the phase, to all atoms of that species.

II. THE ROTATION ACTS ON λ ONLY

The truncated singular vectors are stored in a fixed matrix $B_s^{(\varepsilon)} \in \mathbb{C}^{n_\lambda \times k_s}$, of which the relevant fact is its index structure: it carries an old channel index λ and a reduced channel index $\tilde{\lambda}$, and *nothing else*. Concretely, $B_s^{(\varepsilon)}$ does not depend on

1. the atom a (only on the species s);
2. the transferred momentum $\mathbf{q}$;
3. the reciprocal vector $\mathbf{G}$.

This is the entire content of the separability claim under truncation. Because $B_s^{(\varepsilon)}$ lives in a different index

space from the atom phase and from the K-argument, contraction with $B_s^{(\varepsilon)\dagger}$ commutes with both:

$$\begin{aligned} \sum_{\lambda} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}}^* \left[e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \eta_s(\lambda; \mathbf{q} + \mathbf{G}) \right] \\ = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \underbrace{\sum_{\lambda} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}}^* \eta_s(\lambda; \mathbf{q} + \mathbf{G})}_{\Phi_{\tilde{\lambda}}^{(s)}(\mathbf{q}+\mathbf{G})}. \end{aligned} \quad (2)$$

The right-hand side has the same shape as Eq. (??): an atom phase multiplying an atom-independent K-space function indexed by the reduced channel $\tilde{\lambda}$. This is precisely the definition of the reduced auxiliary in Eq. (eta-reduced-spectral),

$$\eta'_{a\tilde{\lambda}}(\mathbf{G}; \mathbf{q}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \Phi_{\tilde{\lambda}}^{(s)}(\mathbf{q} + \mathbf{G}). \quad (3)$$

No assumption about the SVD itself was used in Eq. (??). For *any* fixed rectangular matrix $B \in \mathbb{C}^{n_\lambda \times k}$, the same identity holds and the same separability follows. The role of the SVD is only to choose a particular B that captures most of the spectrum of η_s at small k .

III. WHERE THE APPROXIMATION ACTUALLY ENTERS

The expansion (eta-low-rank-expansion) of the manuscript states

$$\eta_{a\lambda}^{\mathbf{q}}(\mathbf{G}) \approx \sum_{\tilde{\lambda}} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}} \eta'_{a\tilde{\lambda}}(\mathbf{G}; \mathbf{q}). \quad (4)$$

Substituting Eq. (??) into Eq. (??) and collecting,

$$\sum_{\tilde{\lambda}} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}} \eta'_{a\tilde{\lambda}}(\mathbf{G}; \mathbf{q}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} [B_s^{(\varepsilon)} B_s^{(\varepsilon)\dagger} \eta_s(\cdot; \mathbf{q} + \mathbf{G})]_{\lambda}. \quad (5)$$

Comparing this with Eq. (??) shows that the “ $\approx$ ” in Eq. (??) reduces to a single operation: replacing the full $\eta_s(\lambda; \mathbf{q} + \mathbf{G})$ by its projection onto the column space of $B_s^{(\varepsilon)}$. Two features are notable:

1. the atom phase is identically preserved – it factors out of Eq. (??) with no approximation, exactly as in the original definition;
2. the projector $B_s^{(\varepsilon)} B_s^{(\varepsilon)\dagger}$ acts on the channel index λ only, not on a , $\mathbf{q}$, or $\mathbf{G}$.

In particular, the truncation does not introduce any new dependence on the atom or on $\mathbf{G}$ into the species factor. Whatever separation of indices was present in Eq. (??) is preserved verbatim by Eqs. (??) and (??).

IV. ALGEBRAIC VS. EMPIRICAL CONTENT

It is helpful to keep two layers of the construction distinct, since they are sometimes conflated.

a. Exact (algebraic). For any $n_\lambda \times k$ matrix B , the form $e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \Phi_\lambda^{(s)}(\mathbf{q} + \mathbf{G})$ with $\Phi_\lambda^{(s)}(\mathbf{K}) = \sum_\lambda B_{\lambda\tilde{\lambda}}^* \eta_s(\lambda; \mathbf{K})$ has the same atom-phase / species-resolved separability as the original $\eta_{a\lambda}^{\mathbf{q}}(\mathbf{G})$. This is the algebraic content of Sec. V.A and is independent of how B was chosen.

b. Empirical (numerical). The specific basis $B_s^{(\varepsilon)}$ constructed from a Coulomb-weighted SVD of η_s at $\mathbf{q} = 0$ is observed to cover $\eta_s(\lambda; \mathbf{q} + \mathbf{G})$ at all $\mathbf{q} \neq 0$ with essentially the same projection residual. Equivalently, the column space of $B_s^{(\varepsilon)}$ is closed under the $\mathbf{q}$ -shift in argument. This is the carry-to- $\mathbf{q}$ claim of Sec. V.D. The structural reason is that $\mathbf{q}$ enters only through the magnitude $|\mathbf{q}+\mathbf{G}|$ in the radial Bessel transforms and through the angular factors $Y_{LM}(\widehat{\mathbf{q}+\mathbf{G}})$, both of which vary smoothly within the same finite-dimensional angular-radial subspace as $\mathbf{q}$ sweeps the IBZ. The numerical evidence in Sec. V.D (full $2 \times 2 \times 2$ IBZ for Si PAW/USPP and LiH PAW; residual at $\mathbf{q} \neq 0$ within numerical precision of the residual at $\mathbf{q} = 0$) substantiates this empirically.

The two questions are therefore:

- *Does truncation preserve separability?* Yes, unconditionally and by construction (Eqs. (??)–(??)).
- *Does the $\mathbf{q} = 0$ truncation also approximate η_s at $\mathbf{q} \neq 0$ well?* This is an empirical statement, not an algebraic one, and is the content of the carry-to- $\mathbf{q}$ test in Sec. V.D.

A concern about “separability after truncation” is therefore really either trivial (the algebraic case) or about the rank captured by the $\mathbf{q} = 0$ SVD at finite $\mathbf{q}$ (the empirical case); these are qualitatively different and should be evaluated separately.

V. SEPARABILITY OF THE AUGMENTATION CHARGE

The same separability question can be asked of the augmentation pair density $\hat{Q}_{\alpha\beta}^a$ that drives the local-ISDF construction in Eq. (local-isdf-qhat) of the manuscript. We show that the atom-phase / species-resolved separability of $\hat{Q}$ survives the channel-index truncation by the same mechanism as for η , and discuss the additional structural feature (rank-1 in the channel-pair indices) that makes the THC ansatz hold.

A. The original separable form

The augmentation pair density of atom a , species $s = s(a)$, in Bloch form on the dense $\mathbf{G}$ -grid, is

$$\hat{Q}_{\alpha\beta}^a(\mathbf{G}; \mathbf{q}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} q_{\alpha\beta}^{(s)}(\mathbf{q} + \mathbf{G}), \quad (6)$$

i.e. an atom-dependent unitary phase times the species-resolved augmentation charge $q_{\alpha\beta}^{(s)}(\mathbf{K})$ (the QE $Q^{IJ}(\mathbf{K})$ before structure factor). This is the $\hat{Q}$ analog of Eq. (eta-species-factor) and asserts the same atom-phase / species separation: the atom enters only through the phase and does not act on the channel-pair indices (α, β) .

The full-rank local-ISDF in Eq. (local-isdf-qhat) writes the species augmentation charge as a sum of rank-1 outer products in the channel-pair indices,

$$q_{\alpha\beta}^{(s)}(\mathbf{K}) = \sum_{\lambda=1}^{n_\lambda} U_{\lambda\alpha}^{(s)*} U_{\lambda\beta}^{(s)} \eta_s(\lambda; \mathbf{K}), \quad (7)$$

in the symmetric-pair basis defined by the `build_local_isdf_full_rank` construction ($n_\lambda = n_h^2$; each diagonal pair $I = J$ contributes one row, each off-diagonal pair $I < J$ contributes two rows (λ_+, λ_-) with $U_{\lambda_\pm, I} = +1/\sqrt{2}$, $U_{\lambda_+, J} = +1/\sqrt{2}$, $U_{\lambda_-, J} = -1/\sqrt{2}$, and sign ± 1 absorbed into $\eta_s(\lambda; \cdot)$). At full rank this representation is exact.

B. Atom-phase separability is preserved unconditionally

Combining Eqs. (??)–(??),

$$\hat{Q}_{\alpha\beta}^a(\mathbf{G}; \mathbf{q}) = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \sum_{\lambda} U_{\lambda\alpha}^{(s)*} U_{\lambda\beta}^{(s)} \eta_s(\lambda; \mathbf{q} + \mathbf{G}). \quad (8)$$

The atom enters only through the prefactor; the sum on the right-hand side has no a -dependence beyond $a \mapsto s(a)$. Now apply the truncation argument of Sec. II to the species function under the sum. Replacing $\eta_s(\lambda; \mathbf{K})$ by its rank- k_s projection gives, with $\Pi_s := B_s^{(\varepsilon)} B_s^{(\varepsilon)\dagger}$,

$$\hat{Q}_{\alpha\beta}^a(\mathbf{G}; \mathbf{q}) \approx e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} \sum_{\lambda} U_{\lambda\alpha}^{(s)*} U_{\lambda\beta}^{(s)} [\Pi_s \eta_s(\cdot; \mathbf{q} + \mathbf{G})]_{\lambda}. \quad (9)$$

The atom phase factors out identically: the truncation operator Π_s acts on the channel index λ alone and has no a -, $\mathbf{q}$ -, or $\mathbf{G}$ -dependence. Thus the species-resolved factorization (??) survives – the truncated species charge is still atom-independent, the K-argument still $\mathbf{q} + \mathbf{G}$, and the atom dependence still lives entirely in the phase. This is the augmentation-charge analog of the result in Sec. II–III for η .

C. The rank-1 channel-pair structure

The $\hat{Q}$ factorization in Eq. (??) has an additional structural property that the THC ansatz of Eq. (final-paw-isdf-thc) requires: the coefficient at each channel row λ is a *rank-1 outer product* in the channel-pair indices, $U_{\lambda\alpha}^{(s)*} U_{\lambda\beta}^{(s)}$. This is what permits the linear-in-projector feature $Y_{a\lambda,i} = \sum_{\alpha} U_{a,\lambda\alpha} P_{i,a\alpha}$ of Eq. (Y-feature), and hence the $X_{\Lambda i}^* X_{\Lambda j}$ form of the smooth-grid pair-density factorization in Eq. (comp-rho-local-isdf).

Two compression strategies are visible in the present codebase and manuscript. They differ in how the rank-1 structure interacts with the channel-index truncation, and it is worth being explicit about this distinction.

1. Strategy A – pair-subset selection (current implementation)

The implementation in `local_isdf_compress.hpp` performs a pivoted Cholesky on the pair-space Gram matrix in the L^2 or Coulomb metric and *selects a subset of the original symmetric pairs*. Each kept pair contributes its original symmetric-pair rows (one diagonal row, or a $\pm$ split) verbatim from the full-rank construction. The compressed U is U restricted to those rows; the compressed η_s is η_s restricted to the same rows. The augmentation-charge representation in this case is

$$q_{\alpha\beta}^{(s)}(\mathbf{K}) \approx \sum_{\lambda \in \mathcal{P}_s} U_{\lambda\alpha}^{(s)*} U_{\lambda\beta}^{(s)} \eta_s(\lambda; \mathbf{K}), \quad (10)$$

where $\mathcal{P}_s$ is the kept pair set. The rank-1 outer-product structure in (α, β) is preserved trivially – the truncation discards entire λ rows, and what remains is a direct subset of the original sum. The K_a closed form (`local_isdf.hpp::compute_K_a`) inherits this structure: the round-trip identity $\sum_{\lambda} \text{sign}(\lambda) U_{\lambda I} U_{\lambda J} f(i(\lambda), j(\lambda)) = f(I, J)$ holds restricted to $(I, J) \in \mathcal{P}_s$, and ΔC contributions from dropped pairs are reconstructed as zero. Atom-phase separability and rank-1 channel-pair structure are both exact, with truncation error governed by the pivoted-Cholesky residual in the chosen metric.

2. Strategy B – channel-index SVD rotation (manuscript Sec. V.B): numerical refutation

The manuscript Sec. V.B describes a different compression: rotate the channel index by the truncated SVD basis, $U'_{a,\tilde{\lambda}\alpha} = \sum_{\lambda} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}}^* U_{a,\lambda\alpha}$ (Eq. (U-reduced)), and $\Phi_{\tilde{\lambda}}^{(s)}(\mathbf{K}) = \sum_{\lambda} [B_s^{(\varepsilon)}]_{\lambda\tilde{\lambda}}^* \eta_s(\lambda; \mathbf{K})$ (Eq. (phi-species)). The implicit claim is that the augmentation charge can then be reconstructed in the rotated basis as

$$q_{\alpha\beta}^{(s),B}(\mathbf{K}) := \sum_{\tilde{\lambda}} U'_{\tilde{\lambda}\alpha}{}^{(s)*} U'_{\tilde{\lambda}\beta}{}^{(s)} \Phi_{\tilde{\lambda}}^{(s)}(\mathbf{K}), \quad (11)$$

parallel to Eq. (??). This claim is *numerically false*, and not by a small amount. Substituting the rotation definitions on the right-hand side gives

$$q_{\alpha\beta}^{(s),B}(\mathbf{K}) = \sum_{\lambda_1 \lambda_2 \lambda_3} T_{\lambda_1 \lambda_2 \lambda_3} U_{\lambda_1 \alpha}^{(s)*} U_{\lambda_2 \beta}^{(s)} \eta_s(\lambda_3; \mathbf{K}), \quad (12)$$

with the trilinear kernel

$$T_{\lambda_1 \lambda_2 \lambda_3} = \sum_{\tilde{\lambda}} [B_s^{(\varepsilon)}]_{\lambda_1 \tilde{\lambda}} [B_s^{(\varepsilon)}]_{\lambda_2 \tilde{\lambda}}^* [B_s^{(\varepsilon)}]_{\lambda_3 \tilde{\lambda}}^*. \quad (13)$$

For Eq. (??) to reduce to the original full-rank identity Eq. (??), one would need $T_{\lambda_1 \lambda_2 \lambda_3} = \delta_{\lambda_1 \lambda_2} \delta_{\lambda_1 \lambda_3}$, which holds for $B_s = I$ but *not* for any other unitary – in particular not for the SVD basis of η_s .

The diagnostic test `paw_qhat_strategy_b` (`src/hamiltonian/tests/test_hamilt.cpp`) builds $B_s^{(\varepsilon)}$ from the full Coulomb-weighted SVD of η_s at $\mathbf{q} = 0$, reconstructs $q_{\alpha\beta}^{(s),B}(\mathbf{K})$ per Eq. (??), and compares it to the $\text{qgm}_{\alpha\beta}(\mathbf{K})$ stored in the pseudopotential. Reporting the relative Coulomb-norm residual $\|q^{(s),B} - \text{qgm}\|_C / \|\text{qgm}\|_C$:

species	n_{λ}	full-rank id. $k = n_{\lambda}$ (no trunc.)	$k(10^{-3})$	$k(10^{-9})$
LiH Li PAW	64	4×10^{-19}	1.97	1.97 1.97
LiH H PAW	4	1×10^{-20}	2.61	2.61 2.61
Si PAW	324	5×10^{-20}	1.93	1.93 1.93

The middle column verifies that $\sum_{\lambda} U_{\lambda I} U_{\lambda J} \eta_{\lambda} = \text{qgm}_{IJ}$ holds at machine precision in the symmetric-pair basis – the original full-rank ISDF identity is exact. The remaining three columns show that the Strategy B reconstruction is off by an $\mathcal{O}(1)$ factor at every truncation rank, including the full rank $k = n_{\lambda}$ where the SVD basis itself introduces *no* approximation. The error is the trilinear obstruction Eq. (??), not an SVD truncation residual: the SVD basis cannot make $\sum_{\lambda} B B^* B^*$ into $\delta\delta$, and the manuscript's substitution $\hat{Q} \rightarrow \sum_{\tilde{\lambda}} U'^* U' \Phi$ is therefore *wrong as a representation of the augmentation charge*.

The same conclusion applies to $\Delta C_{a,\alpha\beta,\gamma\delta}$ in Eq. (dC-reduced) of the manuscript: the substitution of $U^* U \rightarrow U'^* U'$ and $K \rightarrow K' = B^\dagger K B$ into Eq. (local-k-factorization) does not reproduce the original ΔC under generic B_s , by the same trilinear obstruction acting on each pair of (α, β) and (γ, δ) indices.

3. Summary

Atom-phase separability of $\hat{Q}$ is preserved unconditionally by either compression strategy, by the argument in Secs. II–III: the truncation acts on an index that is absent from both the atom phase and the K-argument, so the factorization $\hat{Q}^a = e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_a} q^{(s)}$ survives unchanged and the truncated $q^{(s)}$ remains atom-independent.

The rank-1 outer-product structure in (α, β) , however, *cannot* be carried through a generic SVD rotation: $\tilde{C}_{\tilde{\lambda}, \alpha\beta}^{(s)} = \sum_{\lambda} U_{\lambda\alpha}^* U_{\lambda\beta} B_{\lambda\tilde{\lambda}}$ is a weighted sum of rank-1 outer products and is generically of higher rank in (α, β) . Replacing $\tilde{C}^{(s)}$ by $U'^{(s)*} U'^{(s)}$ (Strategy B as written in the manuscript) introduces an $O(1)$ error, demonstrated numerically above.

The implementation in `local_isdf_compress.hpp` avoids this problem by selecting a *subset* of the original symmetric-pair rows (Strategy A): the kept rows retain the rank-1 form verbatim, the roundtrip identity holds restricted to kept pairs, and the truncation error is governed by the pivoted-Cholesky residual in the chosen metric – a single, well-controlled approximation. This is the strategy that should be referenced in Sec. V.B; the SVD-rotation language as currently written in the manuscript is incorrect for $\hat{Q}$ and ΔC , even though it is correct for the species function η_s on its own (Secs. II–III of this note).

VI. NUMERICAL SEARCH FOR TIGHTER COMPRESSIONS

Strategy A preserves the rank-1 form by construction but is restricted to a *subset* of the original $n_{\lambda} = n_h^2$ channel rows. A natural question is whether a different rank-1 outer-product decomposition can do better – i.e. whether

$$q_{\alpha\beta}^{(s)}(\mathbf{K}) \approx \sum_{k=1}^K u_{k\alpha} u_{k\beta} v_k(\mathbf{K}), \quad K \ll n_h^2, \quad (14)$$

admits a small K for given Coulomb-norm tolerance. Equation (??) is the symmetric tensor-rank (CP) decomposition of $q_{\alpha\beta}^{(s)}(\mathbf{K})$ viewed as a complex symmetric 3-tensor in $(\alpha, \beta, \mathbf{K})$, and finding the minimal K is in general NP-hard. Two heuristic constructions were attempted (`paw_qhat_greedy_rank1` and `paw_qhat_outer_svd_inner_takagi` in `src/hamiltonian/tests/test_hamilt.cpp`); both fall short.

A. Greedy rank-1 ALS

The greedy approach extracts a sequence of rank-1 components by power-iteration on Takagi (best symmetric rank-1 fit), subtracts each from the residual, and repeats:

1. Initialize $u \in \mathbb{C}^{n_h}$ (unit norm).
2. $v_g = \sum_{\alpha\beta} \bar{u}_{\alpha} \bar{u}_{\beta} R_{\alpha\beta g}$.
3. $M_{\alpha\gamma} = \sum_g w_C(g) v_g^* R_{\alpha\gamma g}$.
4. $u \rightarrow M\bar{u}/\|M\bar{u}\|$. Iterate until converged.
5. $R \leftarrow R - u u^T \otimes v$; next k .

For Si PAW ($n_h = 18$, $n_{\lambda} = 324$) the greedy residual is 8.6×10^{-2} at $K = 100$, and never reaches 10^{-3} within the $K \leq 100$ budget. For LiH Li PAW ($n_h = 8$, $n_{\lambda} = 64$), $K = 64$ reaches only 5×10^{-4} – well short of machine precision, even though the original symmetric-pair construction is exact at the same K . Greedy CP is famously trapped by local minima; the leading rank-1 components are correct, but later ones must fit a deflated residual that is no longer well-aligned with any single rank-1 outer product.

B. Outer SVD + inner Takagi

A constructive 2-level decomposition truncates first in the $(\alpha\beta, g)$ Coulomb-weighted unfolding, then Takagi-decomposes each leading symmetric $n_h \times n_h$ component:

$$q_{\alpha\beta}^{(s)}(g) \approx \sum_{k=1}^{K_{\text{out}}} \sigma_k W_k(\alpha, \beta) V_k(g), \quad W_k = \sum_{m=1}^{r_k} \mu_m^{(k)} w_m^{(k)} w_m^{(k)T}, \quad (15)$$

giving a total of $\sum_k r_k$ rank-1 outer-product terms. In the test on Si PAW, the leading W_k matrices are essentially full inner rank (typical $r_k(10^{-6}) \sim 12$ –18 out of $n_h = 18$), so the 2-level construction *inflates* the auxiliary dimension well above n_{λ} :

ε_{out}	inner $\varepsilon = 10^{-2} \quad 10^{-3} \quad 10^{-6}$		
LiH Li ($n_{\lambda} = 64$, Coul)			
10^{-3}	57	73	86
10^{-6}	134	166	186
10^{-9}	214	246	266
Si PAW ($n_{\lambda} = 324$, Coul)			
10^{-3}	619	684	842
10^{-6}	1033	1186	1419
10^{-9}	2415	2614	2981

Even the loosest setting ($\varepsilon_{\text{out}} = 10^{-3}$, $\varepsilon_{\text{in}} = 10^{-2}$) gives $619 > n_{\lambda} = 324$ for Si. The leading symmetric directions in the $(\alpha\beta, g)$ unfolding are simply not aligned with single rank-1 outer products: each W_k is a generic complex symmetric matrix that requires $\mathcal{O}(n_h)$ Takagi modes to represent.

C. What this means

These experiments suggest that the symmetric tensor rank of $q_{\alpha\beta}^{(s)}(\mathbf{K})$ is essentially tight at $n_{\lambda} = n_h^2$ under the

rank-1-per-term constraint required by the THC ansatz. The η_s Coulomb-rank diagnostic of Sec. V.A–V.D in the manuscript (Si: $k_\eta \sim 100$ at $\varepsilon = 10^{-6}$) measures the rank of the species function $\eta_s(\lambda; \mathbf{K})$ as a 2-tensor in $(\lambda, \mathbf{K})$. That rank is small. But coupling η_s to the symmetric-pair structure factor $U_{\lambda\alpha}U_{\lambda\beta}$ requires n_h^2 rank-1 outer products in (α, β) , one per λ , and the empirical evidence is that no other rank-1 basis of similar size does any better.

The practical conclusion: Strategy A (pivoted Cholesky pair-subset on the symmetric-pair Gram in the chosen metric, as already implemented in `local_isdf_compress.hpp`) is the right route. It respects the rank-1-per-term structure exactly, achieves the information-theoretic lower bound for that ansatz class, and is controlled by a single tolerance. The Phase 4.5 plan should be updated to describe Strategy A explicitly and drop the B_s -rotation language entirely.

If genuinely smaller compression is required, three escape routes are worth flagging, none free:

1. Replace the symmetric form $u_{k\alpha}u_{k\beta}v_k(g)$ by a non-symmetric form $p_{k\alpha}q_{k\beta}v_k(g)$ (different left and right factors). This breaks the THC $X = Y^*Y$ ansatz and forces a generalized THC kernel.
2. Allow simultaneous (non-greedy) ALS on Eq. (??) with multiple random restarts. This is expensive and offers no rank guarantees – but the symmetric tensor rank may be smaller than the upper bounds the simple heuristics above provide.
3. Exploit physical structure further: angular-momentum coupling selection rules (Wigner 3-j) restrict which (α, β) pairs appear at given (L, M) channels of η_s , suggesting a block-diagonal structure that Strategy A does not exploit.

VII. THE LOCAL COULOMB TENSOR ΔC : SAME STORY

The same compression question can be asked of the one-center correction $\Delta C_{a,\alpha\beta,\gamma\delta}$, which the manuscript factors in the same channel basis as

$$\Delta C_{a,\alpha\beta,\gamma\delta} = \sum_{\lambda\xi} U_{\lambda\alpha} U_{\lambda\beta} K_{a,\lambda\xi} U_{\xi\gamma} U_{\xi\delta}, \quad (16)$$

with K_a the $n_\lambda \times n_\lambda$ real symmetric Coulomb kernel. Two perspectives on compressibility, again testing the THC form against an unconstrained matrix-rank bound.

A. Matrix rank of K_a

The eigendecomposition $K_a = \sum_k \sigma_k v_k v_k^T$ gives the optimal rank- r approximation of the unfolding $\Delta C_{(\alpha\beta),(\gamma\delta)}$. Because ΔC is the difference between AE

species	n_λ	$r(10^{-3})$	$r(10^{-6})$	$r(10^{-9})$	signs (+/-/0)
LiH Li	64	5	15	23	16 / 17 / 31
Si PAW	324	60	97	113	87 / 76 / 161

and PS Coulomb tensors, K_a is generically *indefinite* (signed eigenvalues). Test `paw_K_a_compressibility`:

For Si PAW, half the eigenvalues are zero (K_a has effective rank ≤ 163), and at relative tolerance 10^{-3} the matrix rank is 60 – a $5.4\times$ compression of the unfolding.

B. Constrained ALS: rank-1 form $\sum u_\alpha u_\beta v_\gamma v_\delta$

To compress ΔC while keeping rank-1 outer products in each pair (the THC-preserving constraint, analogous to Eq. (??) for $q^{(s)}$) we ran greedy rank-1 ALS on

$$\Delta C_{\alpha\beta\gamma\delta} \approx \sum_{k=1}^K u_{k\alpha} u_{k\beta} v_{k\gamma} v_{k\delta}. \quad (17)$$

Test `paw_deltaC_greedy_rank1`. Convergence is much slower than the unconstrained matrix-rank bound suggests:

species	n_λ	ALS rank @ 10^{-3}	residual @ $K = 200$
LiH Li	64	28	–
Si PAW	324	<i>not reached</i>	7×10^{-2}

For Si PAW, even $K = 200$ terms in the constrained ALS form reach only $\sim 7 \times 10^{-2}$ relative Frobenius residual – compared to the unconstrained matrix rank 60 at 10^{-3} and rank 97 at 10^{-6} from the eigendecomposition.

C. Why the constrained form costs so much more

The eigendecomposition of K_a writes $\Delta C = \sum_k \sigma_k W_k(\alpha, \beta) W_k(\gamma, \delta)$ with each $W_k(\alpha, \beta) = \sum_\lambda v_{k\lambda} U_{\lambda\alpha} U_{\lambda\beta}$ a generic symmetric $n_h \times n_h$ matrix – not necessarily rank-1. Forcing every term into the form $u_\alpha u_\beta$ instead requires further Takagi-decomposing each W_k into up to n_h rank-1 outer products, multiplying the total auxiliary count by a factor of order n_h . The outer-SVD + inner-Takagi obstruction of Sec. ?? reappears here in exactly the same form.

D. Conclusion: same trilinear obstruction, different tensor

The local Coulomb tensor exhibits the same pattern as the augmentation charge:

- As a generic matrix in $(\alpha\beta, \gamma\delta)$ the unfolding is genuinely low rank (60 of 324 for Si at 10^{-3}).

- Under the rank-1-per-term constraint required by the THC X^*X ansatz, that low rank cannot be exploited – greedy ALS plateaus and the constrained tensor rank is much higher.
- Using the matrix-rank compression directly (rotating to the K_a eigenbasis and identifying $\tilde{\lambda} \rightarrow k$) breaks the THC ansatz by exactly the same trilinear obstruction analyzed in Sec. ?? for $\hat{Q}$: the auxiliary-rotated expression $\sum_k \tilde{U}_k \tilde{U}_k(\sigma_k) \tilde{U}_k \tilde{U}_k$ involves $\sum_{\lambda} v_{k\lambda} v_{k\lambda'} v_{k\lambda''}$, which does not reduce to a Kronecker-delta product.

For Phase 4.5 of the implementation, this leaves Strategy A (pivoted Cholesky pair-subset on the symmetric-pair Gram) as the only THC-respecting compression of either $\hat{Q}$ or ΔC . A genuinely lower-dimensional THC representation of the local block would require giving up the X^*X ansatz on the local rows, e.g. a generalized factorization $\sum_k X_{ki}^Y X_{kj}^Z$ with two distinct features per local row.

VIII. REDEFINING THE AUGMENTATION BY MOMENT-MATCHING ONLY

A different way out of the trilinear wall is to relax the augmentation function itself. The PAW augmentation $\hat{Q}_{\alpha\beta}^a(\mathbf{r})$ is constrained *only* by its multipole moments

$$q_{\alpha\beta}^{LM} = \int d^3r r^L Y_{LM}(\hat{r}) [\rho_{\alpha\beta}^{a,AE}(\mathbf{r}) - \rho_{\alpha\beta}^{a,PS}(\mathbf{r})], \quad (18)$$

which is a *finite* set of linear conditions; the radial shape of $\hat{Q}$ is otherwise free. QE’s canonical construction

$$\hat{Q}_{\alpha\beta}^a(\mathbf{r}) = \sum_{LM} a_{\alpha\beta}^{LM} q_L^{l_{\alpha}l_{\beta}}(|\mathbf{r}|) Y_{LM}(\hat{r}) \quad (19)$$

is one such choice, with $a_{\alpha\beta}^{LM}$ the angular Wigner coupling and $q_L^{l_{\alpha}l_{\beta}}(r)$ the radial Q function.

If one postulates instead a strictly separable form

$$\hat{Q}_{\alpha\beta}^a(\mathbf{r}) \approx \sum_{k=1}^K u_{k\alpha} u_{k\beta} f_k(\mathbf{r}), \quad (20)$$

with f_k any localized shape, the moment-matching condition becomes a constraint on the moments μ_k^{LM} of f_k :

$$q_{\alpha\beta}^{LM} = \sum_k u_{k\alpha} u_{k\beta} \mu_k^{LM}. \quad (21)$$

Since $f_k(\mathbf{r})$ is free up to its moments, the μ_k^{LM} are free parameters; the entire problem reduces to a symmetric CP decomposition of the *moment tensor* $q_{\alpha\beta}^{LM}$, which lives on a tiny finite-dimensional auxiliary index (LM) – not the dense G-grid where the trilinear obstruction was so painful.

A. Numerical test on the moment tensor

Test `paw_moment_separable` builds $q_{\alpha\beta}^{LM} = a_{\alpha\beta}^{LM} \cdot \text{augmom}(L, \beta_{\alpha}, \beta_{\beta})$ from QE’s `ap` table and per-species `augmom` radial moments, then reports two ranks: (i) the *matrix unfolding* rank in $(\alpha\beta\text{-pair}, LM)$, which is the rank achievable in the more flexible Tucker form, and (ii) the symmetric CP rank from greedy rank-1 ALS of Eq. (??).

species	n_h	n_{λ}	n_{LM}	matrix @10 ⁻⁶	matrix @10 ⁻⁹	ALS @10 ⁻³
LiH Li	8	64	9	9	25	34
LiH H	2	4	9	1	3	1
Si PAW	18	324	25	25	130	not reached

The matrix unfolding rank is the rank achievable in the *Tucker* form

$$q_{\alpha\beta}^{LM} = \sum_{k=1}^{K_{\text{Tuck}}} W_k(\alpha, \beta) \nu_k^{LM}, \quad (22)$$

where each W_k is a generic symmetric matrix in (α, β) – not rank-1. For Si this drops the auxiliary count from $n_{\lambda} = 324$ to **25** at 10⁻⁶ tolerance, a 13× reduction. This is the lower bound on what any rank-1-respecting construction can achieve, and it is genuinely small.

The greedy CP rank, however, does not reach this lower bound – for Si, $K = 200$ terms in Eq. (??) still leave $\sim 4 \times 10^{-2}$ relative residual. Putting each W_k into rank-1 outer-product form via Takagi would multiply $K_{\text{Tuck}} = 25$ by an inner Takagi rank that is empirically $\mathcal{O}(n_h)$; the total is no smaller than n_{λ} , and the trilinear wall reappears.

B. What this implies for THC

The conceptual answer to “can we redefine $\hat{Q}$ as separable and still match moments” is unambiguously **yes**: the moment condition is finite and linear, and Eq. (??) admits a CP solution at sufficiently large K . The numerical answer about how small K gets divides into two regimes:

1. **Tucker-like form** (Eq. (??) with generic W_k) reaches $K_{\text{Tuck}} = 25$ for Si at 10⁻⁶, 13× below $n_{\lambda} = 324$. This form is *not* compatible with the standard $X = Y^*Y$ THC ansatz: each W_k is not rank-1 in (α, β) .
2. **Strict CP form** (Eq. (??), rank-1 outer product per term) is what THC requires. Greedy ALS for Si cannot reach 10⁻³ within $K \leq 200$ terms; the symmetric CP rank of the moment tensor is empirically much larger than the matrix-unfolding rank, by the same mechanism that defeats every other rank-1-preserving compression we tried.

In short: the moment-matching reformulation does *not* circumvent the trilinear obstruction. It does, however, sharpen the conclusion: the bottleneck is the rank-1-in- (α, β) constraint required by THC, not the radial-shape freedom that the moment-only condition provides. The intrinsic rank of the moment tensor (25 for Si) is already genuinely small; what fails is the further insistence that each contribution carry a rank-1 outer product. Any future low-rank local-block compression for THC would need to relax the $X = Y^*Y$ ansatz on the local rows, not the choice of augmentation function.

C. Asymmetric form: $\sum u_\alpha v_\beta \mu_{LM}$ with $u \neq v$

A natural generalization is to drop the $(\alpha \leftrightarrow \beta)$ symmetry of each rank-1 component:

$$\hat{Q}_{\alpha\beta}^a(\mathbf{r}) \approx \sum_{k=1}^K u_{k\alpha} v_{k\beta} f_k(\mathbf{r}), \quad u_k \neq v_k, \quad (23)$$

which gives the moment-matching condition $q_{\alpha\beta}^{LM} = \sum_k u_{k\alpha} v_{k\beta} \mu_k^{LM}$ – a plain (asymmetric) CP decomposition of the 3-tensor. This form has strictly more freedom than the symmetric one and matches the form underlying the “two distinct features per local row” ($X_{ki}^Y X_{kj}^Z$) generalization of THC.

Test `paw_moment_asymmetric` runs greedy rank-1 ALS for Eq. (??). Comparison of residuals at matched K :

species	form	residual @ $K = 96$	residual @ $K = 200$
Si PAW	symmetric uu	4.22×10^{-2}	–
Si PAW	asymmetric uv	4.55×10^{-2}	7.9×10^{-3}

The two forms behave nearly identically for the first ~ 100 terms; the asymmetric form pulls ahead modestly at higher K , reaching $\sim 8 \times 10^{-3}$ at $K = 200$ where the symmetric form is still at $\sim 4 \times 10^{-2}$. At nei-

ther setting does the asymmetric CP reach the matrix-unfolding lower bound of 25 (Si at 10^{-6}) or even 10^{-3} within $K \leq 200$.

The structural reason is the same as the outer-SVD+-inner-Takagi analysis for $\hat{Q}^{(s)}$ and ΔC : the leading matrix-unfolding singular components $W_k(\alpha, \beta)$ have SVD-rank $\sim n_h$ in (α, β) , so converting them to rank-1 outer products $u_{k\alpha} v_{k\beta}$ – whether symmetric or asymmetric – still costs $\mathcal{O}(n_h)$ terms per leading W_k . Going asymmetric saves a factor of ≤ 2 (since real symmetric eigendecomposition pairs eigenvalues vs. a single asymmetric SVD), but does not unlock the matrix-rank compression.

The bottom line is the same as before: the rank-1-on-each-pair-side constraint is the hard wall, not the symmetry constraint. Going asymmetric helps a little; it does not help enough to make local-block compression below n_λ feasible while keeping a strict THC-like factorization on each local row.

IX. SUGGESTED CLARIFICATION IN THE MANUSCRIPT

To prevent the present source of confusion in a referee report, one sentence inserted immediately after Eq. (eta-reduced-spectral) is sufficient. A possible wording:

We emphasize that $B_s^{(\epsilon)}$ is a fixed $n_\lambda \times k_s$ matrix with no dependence on the atom a , the transferred momentum $\mathbf{q}$, or the reciprocal vector $\mathbf{G}$. The rotation therefore acts on the channel index λ alone and commutes with both the atom phase and the K -argument. The separable structure of Eq. (eta-species-factor) is preserved by construction, and the only approximation in Eq. (eta-low-rank-expansion) is the rank- k_s truncation in the channel index.